Cercopagis pengoi
(fishhook waterflea)
Crustaceans-Cladocerans
Exotic

Collection Info
Point Map
Species Profile
Animated Map

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Cercopagis pengoi

Common name: fishhook waterflea

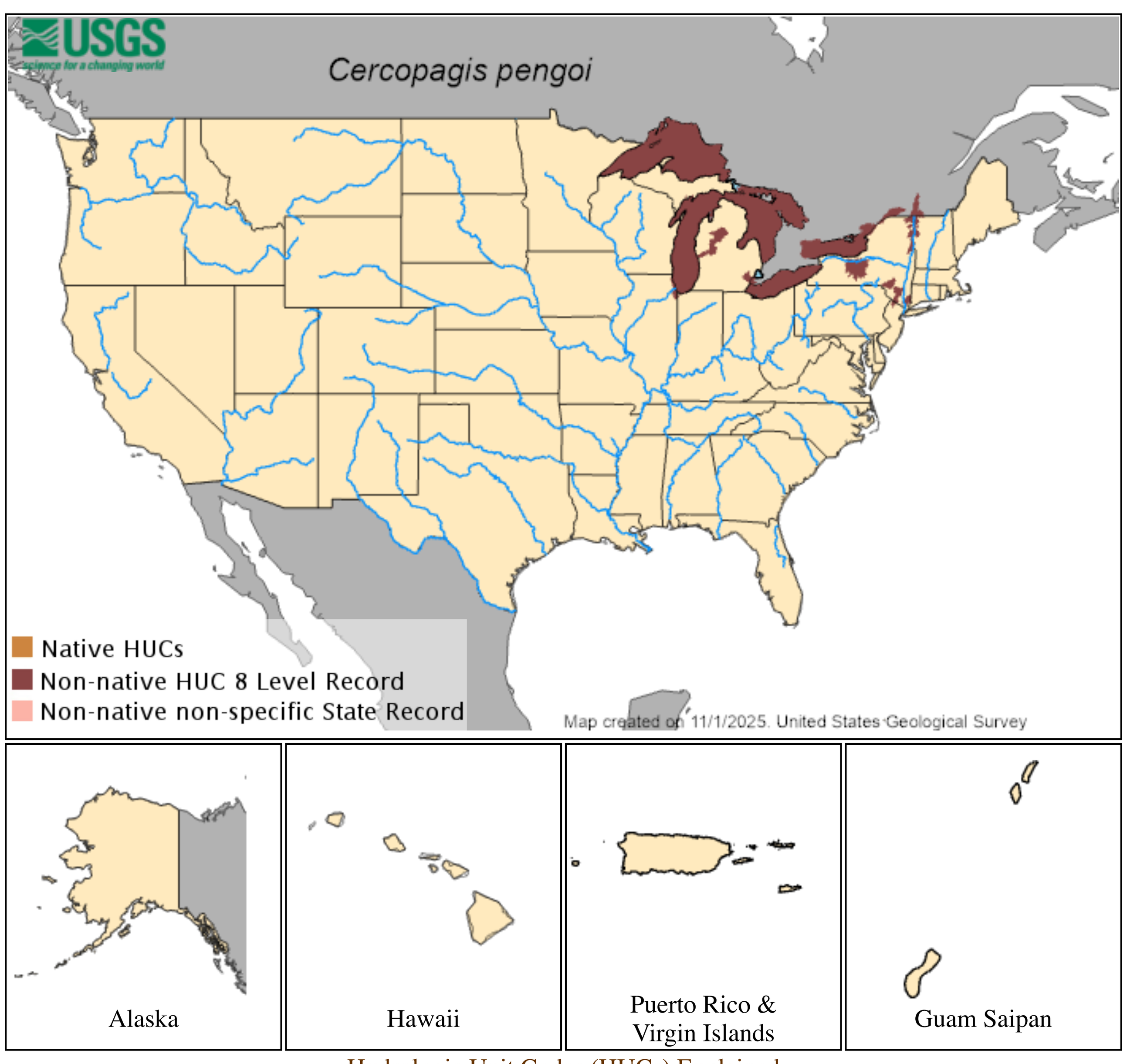
Synonyms and Other Names: fish-hook water flea

Taxonomy: available through ITIS

Identification: Body size from 1–3 mm in length without tail, 6–13 mm with tail; tail has three pairs of barbs and a characteristic loop near the end.

Size: 6 to 13 mm including tail

Native Range: Black, Caspian, Azov, and Aral seas of Europe and Asia (Makarewicz et al. 2001)



Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Cercopagis pengoi* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
IL	1999	2021	2	Chicago; Lake Michigan
MI	1999	2004	5	Detroit; Lake Huron; Lake Michigan; Lake Superior; Muskegon
NY	1998	2024	7	East Branch Delaware; Lake Champlain; Lake Erie; Lake Ontario; Lower Hudson; Rondout; Seneca
OH	2001	2005	1	Lake Erie
PA	2002	2002	1	Lake Erie
WI	2000	2002	1	Lake Michigan

Table last updated 12/3/2025

† Populations may not be currently present.

Ecology: *Cercopagis pengoi* lives in brackish and freshwater lakes. It exhibits diurnal vertical migrations in its native range and feeds on other zooplankton.

In addition to sexual reproduction, *Cercopagis* most commonly reproduces parthenogenically (asexually) during the summer, which allows it to quickly establish new populations with a relatively small seed population without the need for a large number of the smaller males along with females. Eggs produced in the early part of the season are delicate and very susceptible to damage, with low recruitment rates. Later in the season, as surface water temperatures decline, *Cercopagis* females switch to sexual reproduction, producing over-wintering or resting eggs (the species is also known to produce resting eggs anytime during the year when environmental conditions become inhospitable). Such resting eggs can successfully overwinter in an inactive state and replenish the population after hatching in the spring. Resting eggs are also resistant to desiccation, freeze-drying and ingestion by predators (such as other fish). They can be easily transported to other drainage basins by various vectors, particularly if they are still in the female's body (the barbed caudal spine allows attachment to ropes, fishing lines, waterfowl feathers, aquatic gear, vegetation and mud). Resting eggs can hatch regardless of whether the carrier female is alive or dead.

Means of Introduction: Ballast water, boating

Status: Considered established in Lake Ontario, establishing itself quickly (similar to the invasion patterns in Europe) in the other Great Lakes (except L. Huron and Superior) and other inland lakes due to recreational boat traffic and other human activities (USEPA 2008).

Impact of Introduction: *Cercopagis pengoi* is a consumer of other zooplankton. As such it competes with other planktivores of the Great Lakes, including the alewife (*Alosa pseudoharengus*) and rainbow smelt (*Osmerus mordax*) (Bushnoe et al. 2003). Its long spine makes it less palatable to planktivorous fish. For these reasons *C. pengoi* could have a serious effect on the food supply of planktivores. For example, yearling alewife compete directly with *C. pengoi* because they are planktivorous, and cannot consume *C. pengoi* due to the caudal appendage. Once alewife reach their first year they are large enough to handle the caudal appendage (Bushnoe et al. 2003). *Cercopagis pengoi*'s establishment in Lake Ontario in 1998 corresponded with the lowest alewife populations in twenty years (Makarewicz et al. 2001). *Cercopagis pengoi* also fouls fishing gear.

A 2002 study of the food web impacts of *C. pengoi* showed that the depth at which *C. pengoi* exists is depleted of small organisms (<0.15 mg) (Benoit et al. 2002). It is unclear as to whether this is due to predator evasion or *C. pengoi* consumption, but in either case the smaller organisms are forced into deeper, cooler strata, causing growth rate changes (Benoit et al. 2002). The full impact of *C. pengoi* on the food web has not yet been extensively studied.

Remarks: *Cercopagis pengoi* has been found in the stomach of some fishes in high percentages in Europe.

According to the EPA's GARP model, *C. pengoi*, a free-swimming macroinvertebrate, would likely find suitable habitat throughout the Great Lakes region, except for the deeper waters of Lake Superior. However, population densities of the fishhook water flea increase with distance from shore (IUCN 2010), suggesting that this species may be able to occupy the entire region, including the deeper waters of Lake Superior, given sufficient time (USEPA 2008).

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Other Resources:
Army Corps of Engineers, WES

Cercopagis pengoi (ANS Clearinghouse Bibliography)

Cercopagis pengoi (Global Invasive Species Database)

Great Lakes Water Life

New York Invasive Species Clearinghouse

US Fish and Wildlife Service Ecological Risk Screening Summary for *Cercopagis pengoi*

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